

Amplituhedron — Clean Formulation v1

0. Scope

A self-contained definition of the amplituhedron $\mathcal{A}_{k,m,n}(Z)$, its basic properties, and a minimal, correct path to the canonical form. Assumes familiarity with linear algebra and differential forms.

1. Notation and preliminaries

- Work over $\mathbb{R}$. Projective space $\mathbb{P}^r$ is lines through the origin in $\mathbb{R}^{r+1}$.
- The Grassmannian $\text{Gr}(k, N)$ is the set of k -planes in $\mathbb{R}^N$. Points are represented by full-rank $k \times N$ matrices modulo left GL_k .
- Plücker coordinates for $\text{Gr}(k, N)$ are the $k \times k$ minors of a representative matrix. They satisfy the Plücker relations and are defined up to an overall scale.
- The **positive Grassmannian** $G_+(k, n)$ consists of $k \times n$ real matrices of rank k whose maximal ordered minors are all strictly positive, modulo left GL_k .
- Indices are cyclic modulo $n : i \equiv i + n$.

2. External data Z

Let $Z \in \mathbb{R}^{n \times (k+m)}$ have full rank $k + m$. Assume **positivity of Z** : all ordered $(k + m)$ minors of Z are strictly positive. Columns are $Z_1, \dots, Z_n$ with a fixed cyclic order.

Two data sets Z, Z' related by $Z' = Zg$ with $g \in \text{GL}_{k+m}^+$ encode the same amplituhedron as subsets of projective space. We work projectively in $\text{Gr}(k, k + m)$ or $\mathbb{P}^{\binom{k+m}{k}-1}$ as convenient.

3. Definition of the amplituhedron

Let $C \in G_+(k, n)$. Define a map

$$\Phi_Z : G_+(k, n) \rightarrow \text{Gr}(k, k + m), \quad C \mapsto Y = \text{rowspan}(CZ).$$

The **amplituhedron** is the image

$$\mathcal{A}_{k,m,n}(Z) := \{ Y = \text{rowspan}(CZ) : C \in G_+(k, n) \} \subset \text{Gr}(k, k + m).$$

Properties: - $\dim \mathcal{A}_{k,m,n}(Z) = km$. - $\mathcal{A}_{k,m,n}(Z)$ is invariant under left multiplication $C \mapsto gC$ with $g \in \text{GL}_k$ and right multiplication $Z \mapsto Zh$ with $h \in \text{GL}_{k+m}^+$ (projectively). - Boundaries of $\mathcal{A}_{k,m,n}(Z)$ are images of boundaries of positroid cells in $G_+(k, n)$ under Φ_Z .

4. Canonical form

There exists a unique top-degree differential form $\Omega_{k,m,n}(Z)$ on $\mathcal{A}_{k,m,n}(Z)$ characterized by: 1. **Logarithmic singularities only** on all boundary components of $\mathcal{A}_{k,m,n}(Z)$. 2. **Unit leading residues** on each top-dimensional cell in any triangulation of $\mathcal{A}_{k,m,n}(Z)$.

Computation strategy: - Choose a positroid triangulation $\{\mathcal{C}_\alpha\}$ of $G_+(k, n)$ such that Φ_Z is birational on each $\mathcal{C}_\alpha$. - On each $\mathcal{C}_\alpha$, use positive coordinates $(x_1, \dots, x_{km})$ in which the pullback of Ω is a product of $d\log$ factors:

$$\Phi_Z^* \Omega|_{\mathcal{C}_\alpha} = d\log x_1 \wedge \dots \wedge d\log x_{km}.$$

- Pushforward to Y -space and sum over α :

$$\Omega_{k,m,n}(Z) = \sum_{\alpha} (\Phi_Z)_* (d\log x_1 \wedge \dots \wedge d\log x_{km}).$$

Independence of triangulation follows from local compatibility of leading residues on shared boundaries.

5. Examples

5.1 Polygon case: $k = 1, m = 2$

- Data: $Z \in \mathbb{R}^{n \times 3}$ with all ordered 3×3 minors positive. The image $\mathcal{A}_{1,2,n}(Z) \subset \mathbb{P}^2$ is the convex n -gon with vertices given by consecutive pairs $[Z_i Z_{i+1}]$.
- Canonical form: Let a fan triangulation at vertex 1 give triangles $(1, i, i+1)$ for $i = 2, \dots, n-1$. The canonical form is the sum of triangle canonical forms pulled forward from positive coordinates on each cell. Different triangulations produce the same $\Omega_{1,2,n}(Z)$.

5.2 $k = 1, m = 4$

- $\mathcal{A}_{1,4,n}(Z) \subset \mathbb{P}^4$ is a cyclic polytope. The canonical form is computed by any regular triangulation compatible with the cyclic order; it again equals the sum of simplex canonical forms with unit residues.

5.3 Minimal nontrivial $k > 1$

- Take $k = 2, m = 2, n \geq 4$. Then $\dim \mathcal{A} = 4$. Choose a plabic-graph triangulation of $G_+(2, n)$, assign cluster-like positive coordinates to each top cell, evaluate $d\log$ forms, and pushforward.

6. Boundaries and positroid stratification

- Positroid cells in $G_+(k, n)$ are labeled by decorated permutations or plabic graphs. Codimension-one boundaries correspond to the vanishing of a single Plücker coordinate consistent with positivity.
- Under Φ_Z , these cells map to boundaries of $\mathcal{A}_{k,m,n}(Z)$. The canonical form can be reconstructed by requiring that residues on all such boundaries match those induced from lower-dimensional amplituhedra.

7. Symmetries

- Z with ordered positive minors implies invariance under the dihedral action on indices $i \mapsto i \pm 1$ and reversal, realized projectively.
- GL_{k+m}^+ acts by right multiplication on Z , yielding equivalent embeddings of $\mathcal{A}_{k,m,n}$.

8. Algorithms at a glance

1. Fix Z with positive ordered minors.
2. Choose a positroid triangulation of $G_+(k, n)$ (e.g., via a plabic graph).
3. For each top cell, pick positive coordinates making the pullback a dlog volume form.
4. Pushforward through Φ_Z and sum over cells.
5. Verify triangulation-independence by matching residues on shared boundaries.

9. Sanity checks

- **Dimension:** $\dim G_+(k, n) = k(n-k)$. The map Φ_Z has fiber dimension $k(n-k-m)$, hence $\dim \mathcal{A} = km$.
- **Convexity for $k = 1$:** positivity of ordered minors implies the image is a cyclic polytope, thus convex.
- **Projective invariance:** Ω is defined up to overall scale fixed by the unit-residue normalization.

10. Minimal glossary

- **Plücker bracket** $\langle i_1 \cdots i_k \rangle$: the $k \times k$ minor of a matrix formed from columns $i_1, \dots, i_k$.
- **Canonical form:** a differential form with logarithmic singularities on all boundaries and unit residues on top cells; unique on a positive geometry.
- **Positive geometry:** a pair $(X, X_{\geq 0})$ where X is a complex projective variety and $X_{\geq 0} \subset X(\mathbb{R})$ carries a unique canonical form.

11. References for further study

- Positive Grassmannian, plabic graphs, and cluster coordinates.
- Amplituhedron and positive geometries in scattering amplitudes.

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